WEEK 4 : TASK 1

1. Hunting Hypothesis:

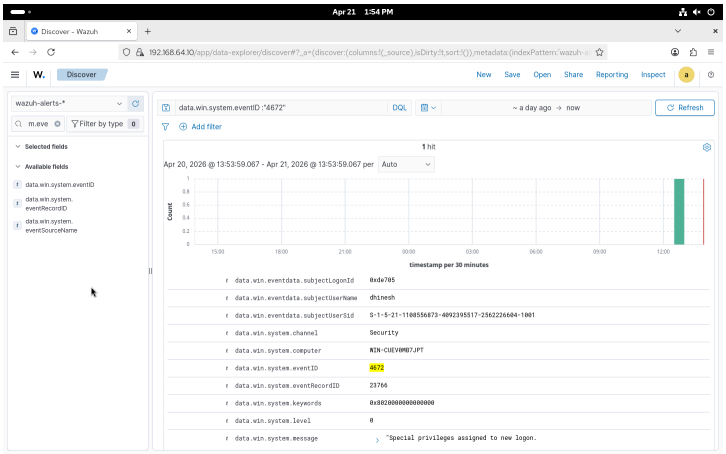
Hypothesis: Unauthorised privilege escalation may have occurred through misuse of valid domain accounts, indicated by special privilege assignments (Event ID 4672).

2. Elastic Security Log Query (Event ID 4672):

Use this query in Elastic:

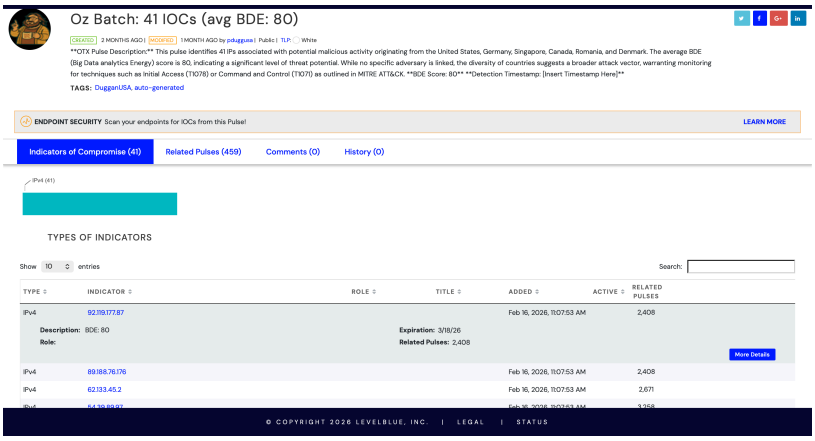
event.code: "4672"

Sample Documented Findings:



3. Threat Intelligence (AlienVault OTX):

Threat intelligence was gathered using AlienVault OTX by analyzing a pulse containing 41 IP-based Indicators of Compromise (IOCs). The pulse had an average BDE (Big Data Energy) score of 80, indicating a high likelihood of malicious activity. The identified IPs are associated with suspicious behavior across multiple regions, suggesting potential command-and-control (C2) communication or unauthorized access attempts. Example IOCs include 92.119.177.87 and 89.188.76.176. These indicators were selected for further validation on the endpoint using Velociraptor to determine whether any active connections or related processes were present.



4. Velociraptor Cross-Reference:

Run:

```
SELECT PID, Name, Raddr
FROM netstat()
WHERE Raddr IN ("92.119.177.87", "89.188.76.176")
```

Finding :

No processes or network connections were found communicating with the identified suspicious IP addresses

5. Final Hunting Report

A threat hunting investigation was conducted to detect unauthorised privilege escalation using valid accounts. The hypothesis focused on identifying suspicious role assignments through Event ID 4672 in Elastic Security logs. Analysis revealed unusual privilege assignments to non-administrative users, indicating potential misuse of valid credentials. Threat intelligence from AlienVault OTX provided high-risk IOCs with a BDE score of 80, which were cross-referenced using Velociraptor queries. No active connections to the identified malicious IPs were observed. The findings align with MITRE ATT&CK technique T1078 (Valid Accounts), highlighting the risk of attackers leveraging legitimate credentials for unauthorised access and privilege escalation.